

Maitreya Mohapatra

AI Engineer

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📅 **Available:** full-time from 24 Aug 2026 🌐 **Portfolio:** mimo1999.github.io/



Professional Summary

AI Engineer (3+ yrs) building production agentic/RAG systems and CI/CD-deployed ML services across fintech, healthcare, and industrial R&D. Co-inventor, 2 US patents. Completing M.Sc. AI at FAU Erlangen-Nürnberg.

Work Experience

AI Infra Engineer (Werkstudent) | Siemens Healthineers | Erlangen, Germany 04/2025 – present

Python, Azure DevOps, Slurm, Podman, Linux, Pytest

- Designed an MCP-based orchestration workflow for Questa OneSpin RTL validation via FuseSoC.
- Redesigned Azure DevOps CI/CD into a parallel execution framework, reducing runtime by ~20% and manual effort by ~60%.
- Architected a hybrid Azure DevOps–Slurm execution pipeline for containerized FPGA simulation, improving compute utilization by ~50% and reducing turnaround by ~35%.

Teaching Assistant, M.Sc. DL | FAU Erlangen-Nürnberg | Erlangen, Germany 11/2024 – 03/2025

PyTorch, Python

- Mentored M.Sc. students in implementing and debugging CNN, RNN, and Transformer models using PyTorch.
- Guided students on optimization, regularization, experiment debugging, and deep learning best practices during practical sessions.

Associate Software Engineer, Data Science | HighRadius | Bhubaneswar, India 06/2021 – 08/2023

Python, SQL, Snowflake, Jenkins, Docker, AWS

- Led ML treasury-forecasting delivery for 8 enterprise accounts as primary technical contact; mentored 10+ interns.
- Built two patented models: payroll cash-flow forecasting (95%+ accuracy) and a customer-behavior feature framework that lifted accuracy 20–30% across production accounts.
- Cut AWS costs ~20% via compute-efficient pipeline design, and client implementation time 10–15% by standardizing multi-ERP (SAP, Oracle, NetSuite) ingestion into SQL/Snowflake ETL pipelines.

Education

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) 10/2023 – 08/2026

Master of Science in Artificial Intelligence (*graduating August 2026*)

Grade: 1.6 (German scale)

Kalinga Institute of Industrial Technology, Bhubaneswar (KIIT) 06/2017 – 06/2021

Bachelor of Technology in Computer Science and Engineering

Grade: 1.5 (German scale equivalent)

Technical Skills

- AI Systems & LLM Engineering:** Agentic & multi-agent systems, MCP, RAG & GraphRAG, tool calling, structured outputs, context engineering, OpenAI/Anthropic APIs, LangGraph, LangChain
- LLM Evaluation & Observability:** Ragas, LangSmith, Langfuse, MLflow, LLM-as-judge, evaluation pipelines, experiment tracking, model evaluation & benchmarking
- AI Infrastructure & Serving:** FastAPI, Docker, Linux, CI/CD (Azure DevOps, Jenkins), Slurm/HPC, model serving, async APIs, AWS/Azure/GCP, on-premise & offline deployment
- Data, Retrieval & ML:** PostgreSQL/pgvector, ChromaDB, Snowflake, Databricks, vector search, hybrid retrieval, reranking, PyTorch, scikit-learn
- Programming:** Python, SQL, C++, TypeScript/JavaScript, Java, Bash

Languages

English (C2) · German (A2, actively improving) · Hindi (C2) · Bengali (C1)

Research, Patents & Projects

- Master's Thesis:** Built a multimodal (HSV+audio) cross-attention TCN for clinical voice diagnostics, achieving 86% patient-level accuracy and integrated into a Python clinical application.
- US Patents (HighRadius):** Cash-flow forecasting (granted, US-12657596-B2); transaction-pattern prediction (applied).
- Projects:** **Clinical RAG:** offline German clinical-guideline RAG (hybrid BM25+dense, reranking); 88.9% Recall@5, 100% refusal correctness out-of-domain; **Research Swarm:** LangGraph 5-agent system with HITL; BEIR nDCG@10 0.749 on SciFact matching a 10× larger embedder, 181 offline tests; **ChemoGAM:** interpretable EBMs vs. CatBoost+SHAP on MIMIC-IV (AUROC 0.826 vs 0.818); shipped **sstgam**, an MIT pip package.
- Awards:** 3rd, Medical Valley Healthcare Hackathon 2024 (patient-facing RAG assistant); 2nd, Siemens × Neo4j Graphathon 2026 (GraphLLM for competitive intelligence).